

Exact Certificates and Active-Set Reductions for Additive Chore Fisher Markets

A rigorous note on the Brânzei–Sandomirskiy open problem

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Abstract

This note does *not* solve the unrestricted polynomial-time exact-computation problem for competitive division of additive chores. It records partial progress in the following precise sense. For the additive divisible-chores Fisher market with strictly negative values and strictly negative budgets, we prove an exact complementarity certificate in prices, earning multipliers, and money flows. Once a candidate active graph is known, the certificate becomes a rational linear feasibility problem, so an exact competitive equilibrium and its utility profile can be recovered in polynomial time. Equivalently, competitive equilibria are exactly the stationary points of a natural nonsmooth difference-of-convex potential after price normalization. These statements give a clean verifier, a finite exact enumeration algorithm, and polynomial-time exact recovery under an active-set oracle or a nondegenerate active-set-identification promise. They do not give a polynomial-time rule for selecting one feasible active graph when both the number of agents and the number of chores vary; that graph-selection step is exactly the unresolved part.

1 Introduction and status of the result

Brânzei and Sandomirskiy study competitive division of divisible chores under additive utilities, where every item is a bad and budgets are liabilities rather than ordinary spending budgets [1]. Their main exact algorithm computes all competitive utility profiles in strongly polynomial time when either the number of agents or the number of chores is fixed. The unrestricted question asks whether, when both sides of the market vary, one can compute at least one competitive utility profile in time polynomial in the input size.

The conclusion of this note is deliberately modest. We do not obtain such a polynomial-time algorithm. Instead, we isolate a short exact certificate and show that the only missing step is active-set selection. In particular:

- (i) A competitive equilibrium is equivalent to a finite complementarity system in prices p , earning multipliers β , and money flows f .
- (ii) If a graph G is supplied that contains the positive-flow support and is contained in the tight graph of some equilibrium, exact recovery is a linear programming problem.
- (iii) Competitive equilibria are also exactly the stationary points of a scale-invariant nonsmooth potential

$$\Phi(\mu) = \sum_i B_i \max_j (\mu_j - \log d_{ij}) - B \log \left(\sum_j e^{\mu_j} \right).$$

The first two points are useful only after the relevant active graph is known or guessed. Enumerating all graphs gives a finite exact algorithm but not a polynomial-time algorithm in the unrestricted case. Thus the contribution should be read as a rigorous reduction and conditional recovery theorem, not as a solution of the open problem.

2 Model

Let $[n] = \{1, \dots, n\}$ be the set of agents and $[m] = \{1, \dots, m\}$ the set of divisible chores. In the usual chores convention, values and budgets are strictly negative. We use the equivalent positive-disutility convention

$$d_{ij} = -v_{ij} > 0, \quad B_i = -b_i > 0,$$

where d_{ij} is the disutility suffered by agent i from one unit of chore j , and B_i is the amount of compensation that agent i must earn. Put

$$B = \sum_{i=1}^n B_i.$$

There is one unit of every chore. An allocation is a matrix $x \in \mathbb{R}_{\geq 0}^{n \times m}$ satisfying

$$\sum_{i=1}^n x_{ij} = 1 \quad (j \in [m]).$$

The original negative utility of agent i is

$$u_i(x) = - \sum_{j=1}^m d_{ij} x_{ij}.$$

Prices are written as positive compensations: performing one unit of chore j earns $p_j > 0$. Given prices $p \in \mathbb{R}_{> 0}^m$, agent i solves

$$\min_{y \in \mathbb{R}_{\geq 0}^m} d_i \cdot y \quad \text{subject to} \quad p \cdot y \geq B_i. \quad (1)$$

Definition 2.1 (Competitive equilibrium). *A pair (x, p) is a competitive equilibrium if $p \in \mathbb{R}_{> 0}^m$, $x \geq 0$, all chores are fully allocated, every agent earns exactly her required compensation,*

$$p \cdot x_i = \sum_{j=1}^m p_j x_{ij} = B_i \quad (i \in [n]),$$

and x_i solves (1) for every agent i .

For later use define the maximum earning-per-disutility ratio of agent i at prices p by

$$\beta_i(p) = \max_{j \in [m]} \frac{p_j}{d_{ij}}, \quad (2)$$

and define the corresponding tight set

$$M_i(p) = \arg \max_{j \in [m]} \frac{p_j}{d_{ij}}. \quad (3)$$

3 The single-agent optimality condition

The following elementary lemma is the source of all complementarity conditions below.

Lemma 3.1 (Agent optimality). *Fix $p \in \mathbb{R}_{>0}^m$ and an agent i . Let $\beta_i = \beta_i(p)$. Then the optimal value of (1) is B_i/β_i . A bundle $y \in \mathbb{R}_{\geq 0}^m$ is optimal if and only if*

$$p \cdot y = B_i \quad \text{and} \quad \text{supp}(y) \subseteq M_i(p).$$

Proof. For every chore j , the definition of β_i gives

$$p_j \leq \beta_i d_{ij}.$$

Hence every feasible bundle y satisfies

$$d_i \cdot y \geq \frac{1}{\beta_i} p \cdot y \geq \frac{B_i}{\beta_i}.$$

Thus no feasible bundle has disutility below B_i/β_i .

Conversely, choose any $j \in M_i(p)$ and set $y_j = B_i/p_j$ and $y_k = 0$ for $k \neq j$. This bundle is feasible, earns exactly B_i , and has disutility

$$d_{ij} \frac{B_i}{p_j} = \frac{B_i}{p_j/d_{ij}} = \frac{B_i}{\beta_i}.$$

Therefore the lower bound is tight and is the optimal value.

The equality conditions in the displayed lower bound are also clear. Equality in $p \cdot y \geq B_i$ requires $p \cdot y = B_i$, and equality in

$$d_i \cdot y \geq \frac{1}{\beta_i} p \cdot y$$

requires $y_j > 0$ only when $p_j = \beta_i d_{ij}$, equivalently $j \in M_i(p)$. This proves the characterization of optimal bundles. $\square$

4 Exact complementarity certificate

We now introduce money-flow variables

$$f_{ij} := p_j x_{ij}.$$

Thus f_{ij} is the amount of compensation earned by agent i from chore j .

Theorem 4.1 (Complementarity certificate). *For strictly positive rational or real data $d \in \mathbb{R}_{>0}^{n \times m}$ and $B \in \mathbb{R}_{>0}^n$, a competitive equilibrium exists with prices p and money flows $f_{ij} = p_j x_{ij}$ if and only if there are variables*

$$p \in \mathbb{R}_{>0}^m, \quad \beta \in \mathbb{R}_{>0}^n, \quad f \in \mathbb{R}_{\geq 0}^{n \times m}$$

satisfying

$$\sum_{j=1}^m f_{ij} = B_i \quad (i \in [n]), \quad (4)$$

$$\sum_{i=1}^n f_{ij} = p_j \quad (j \in [m]), \quad (5)$$

$$p_j \leq \beta_i d_{ij} \quad (i \in [n], j \in [m]), \quad (6)$$

$$f_{ij}(\beta_i d_{ij} - p_j) = 0 \quad (i \in [n], j \in [m]). \quad (7)$$

Given such a certificate, the allocation

$$x_{ij} = \frac{f_{ij}}{p_j} \quad (8)$$

is a competitive equilibrium allocation, and the resulting utility profile is

$$u_i = -\frac{B_i}{\beta_i} \quad (i \in [n]). \quad (9)$$

Proof. Suppose first that (p, β, f) satisfies (4)–(7), and define x by (8). Since $p_j > 0$, this is well defined. By (5), for each chore j ,

$$\sum_{i=1}^n x_{ij} = \frac{1}{p_j} \sum_{i=1}^n f_{ij} = 1,$$

so all chores are fully allocated. By (4), for each agent i ,

$$p \cdot x_i = \sum_{j=1}^m p_j x_{ij} = \sum_{j=1}^m f_{ij} = B_i.$$

The inequalities (6) imply $p_j/d_{ij} \leq \beta_i$ for every chore j . If $x_{ij} > 0$, then $f_{ij} > 0$, and complementarity (7) gives $p_j = \beta_i d_{ij}$. Thus x_i is supported on chores maximizing p_j/d_{ij} . Lemma 3.1 shows that x_i is optimal in the agent problem. Hence (x, p) is a competitive equilibrium.

Moreover, complementarity on the support of f_i gives

$$d_i \cdot x_i = \sum_{j: f_{ij} > 0} d_{ij} \frac{f_{ij}}{p_j} = \sum_{j: f_{ij} > 0} \frac{f_{ij}}{\beta_i} = \frac{B_i}{\beta_i},$$

which proves (9).

Conversely, let (x, p) be a competitive equilibrium. Define

$$\beta_i = \max_j \frac{p_j}{d_{ij}}, \quad f_{ij} = p_j x_{ij}.$$

Budget balance gives (4); full allocation gives (5); and the definition of β_i gives (6). By Lemma 3.1, every optimal bundle x_i is supported on $M_i(p)$. Therefore $f_{ij} > 0$ implies $p_j = \beta_i d_{ij}$, which is exactly (7). This completes the proof. $\square$

Remark 4.2 (Verification versus computation). *Theorem 4.1 gives an exact polynomial-time verifier for an explicitly supplied rational certificate (p, β, f) . It does not, by itself, give a polynomial-time algorithm for finding such a certificate. The difficulty is the complementarity condition (7): one must know which zero products are enforced by zero flow and which are enforced by tight ratios.*

5 Linear recovery from an active graph

The complementarity products become linear once the relevant active graph is supplied. For a certificate (p, β, f) define its positive-flow support and tight graph by

$$S(f) = \{(i, j) : f_{ij} > 0\}, \quad T(p, \beta) = \{(i, j) : p_j = \beta_i d_{ij}\}.$$

The certificate conditions imply

$$S(f) \subseteq T(p, \beta).$$

Thus the useful object is not necessarily the exact support. It is any graph G satisfying

$$S(f) \subseteq G \subseteq T(p, \beta)$$

for some equilibrium certificate.

Fix a proposed graph $G \subseteq [n] \times [m]$. Consider the following linear program, denoted $\text{LP}(G)$:

$$\begin{aligned} & \text{maximize} \quad \tau \\ & \text{subject to} \quad \tau \geq 0, \quad p_j \geq \tau(j \in [m]), \quad \beta_i \geq \tau(i \in [n]), \end{aligned} \tag{10}$$

$$f_{ij} \geq 0 \quad (i \in [n], j \in [m]), \tag{11}$$

$$f_{ij} = 0 \quad ((i, j) \notin G), \tag{12}$$

$$\sum_{j=1}^m f_{ij} = B_i \quad (i \in [n]), \tag{13}$$

$$\sum_{i=1}^n f_{ij} = p_j \quad (j \in [m]), \tag{14}$$

$$p_j = \beta_i d_{ij} \quad ((i, j) \in G), \tag{15}$$

$$p_j \leq \beta_i d_{ij} \quad (i \in [n], j \in [m]). \tag{16}$$

All constraints are linear in (τ, p, β, f) .

Proposition 5.1 (Graph-conditioned exact recovery). *Assume d and B are rational. Given a graph $G \subseteq [n] \times [m]$, one can decide in time polynomial in n , m , and the input bit length whether $\text{LP}(G)$ has optimum value $\tau^* > 0$. If $\tau^* > 0$, then any optimal solution gives an exact rational competitive equilibrium by*

$$x_{ij} = \frac{f_{ij}}{p_j}, \quad u_i = -\frac{B_i}{\beta_i}.$$

Conversely, if there is an equilibrium certificate (p, β, f) with

$$S(f) \subseteq G \subseteq T(p, \beta),$$

then $\text{LP}(G)$ has optimum value $\tau^* > 0$.

Proof. The program is a rational linear program of size polynomial in nm and in the bit length of the data. Standard polynomial-time linear programming algorithms decide feasibility and optimization, and an exact rational basic optimal solution can be recovered with polynomial bit length.

If $\tau^* > 0$, then $p_j > 0$ and $\beta_i > 0$. Conditions (13), (14), and (16) are exactly (4), (5), and (6). For complementarity, if $(i, j) \notin G$ then $f_{ij} = 0$ by (12); if $(i, j) \in G$ then $p_j = \beta_i d_{ij}$ by (15). Hence $f_{ij}(\beta_i d_{ij} - p_j) = 0$ for all i, j . Theorem 4.1 then gives a competitive equilibrium and the stated utility formula.

Conversely, suppose an equilibrium certificate (p, β, f) satisfies $S(f) \subseteq G \subseteq T(p, \beta)$. Then $f_{ij} = 0$ outside G , all edges of G are tight, and the certificate satisfies all other linear constraints. Since p and β are strictly positive, setting

$$\tau = \min\left\{\min_j p_j, \min_i \beta_i\right\} > 0$$

makes the solution feasible for $\text{LP}(G)$ with positive objective value. Therefore $\tau^* > 0$. $\square$

Corollary 5.2 (Finite exact enumeration). *Enumerating all graphs $G \subseteq [n] \times [m]$ and applying Proposition 5.1 gives a finite exact algorithm for finding a competitive equilibrium. The running time is exponential in nm in the worst case.*

Corollary 5.3 (Conditional polynomial-time recovery). *Suppose that, for every instance, one can construct in polynomial time a family $\mathcal{G}$ of graphs of polynomial size such that at least one $G \in \mathcal{G}$ satisfies*

$$S(f) \subseteq G \subseteq T(p, \beta)$$

for some equilibrium certificate. Then an exact competitive equilibrium and utility profile can be computed in polynomial time.

Proof. Run $\text{LP}(G)$ for every $G \in \mathcal{G}$ and output the first graph with $\tau^* > 0$. Proposition 5.1 proves correctness and polynomial running time under the stated hypothesis. $\square$

Remark 5.4 (The unresolved step). *Corollary 5.3 is only conditional. Brânzei and Sandomirskiy construct polynomial-size graph families when one side of the market is fixed [1]. The unrestricted open problem asks, in effect, whether one can avoid exponential active-graph search when both n and m vary.*

6 A nonsmooth stationary formulation

The same certificate can be written as a stationary condition for a scale-invariant nonsmooth potential. For $\mu \in \mathbb{R}^m$, define

$$h_i(\mu) = \max_{j \in [m]} (\mu_j - \log d_{ij})$$

and

$$\Phi(\mu) = \sum_{i=1}^n B_i h_i(\mu) - B \log \left(\sum_{j=1}^m e^{\mu_j} \right). \quad (17)$$

The function is invariant under adding a constant to all coordinates of μ . For each i , set

$$M_i(\mu) = \arg \max_{j \in [m]} (\mu_j - \log d_{ij}) = \arg \max_{j \in [m]} \frac{e^{\mu_j}}{d_{ij}}.$$

For a nonempty set $A \subseteq [m]$, write

$$\Delta(A) = \{\lambda \in \mathbb{R}_{\geq 0}^m : \sum_j \lambda_j = 1, \text{ supp}(\lambda) \subseteq A\}.$$

Since h_i is the maximum of finitely many affine functions,

$$\partial h_i(\mu) = \Delta(M_i(\mu)). \quad (18)$$

We use the standard difference-of-convex stationarity condition

$$0 \in \sum_{i=1}^n B_i \partial h_i(\mu) - \nabla \left[B \log \left(\sum_{j=1}^m e^{\mu_j} \right) \right]. \quad (19)$$

Theorem 6.1 (Stationarity equals equilibrium). *A vector $\mu \in \mathbb{R}^m$ satisfies the stationarity condition (19) if and only if the normalized prices*

$$p_j = \frac{Be^{\mu_j}}{\sum_{k=1}^m e^{\mu_k}} \quad (20)$$

can be extended to a competitive equilibrium. More explicitly, stationarity is equivalent to the existence of vectors

$$\lambda_i \in \Delta(M_i(\mu)) \quad (i \in [n])$$

such that

$$\sum_{i=1}^n B_i \lambda_{ij} = p_j \quad (j \in [m]). \quad (21)$$

Given such vectors, the money flows

$$f_{ij} = B_i \lambda_{ij}$$

and the allocation $x_{ij} = f_{ij}/p_j$ form a competitive equilibrium.

Proof. The gradient of the smooth term is

$$\nabla_j \left[B \log \left(\sum_{k=1}^m e^{\mu_k} \right) \right] = \frac{Be^{\mu_j}}{\sum_{k=1}^m e^{\mu_k}} = p_j.$$

Together with (18), the stationarity condition (19) is exactly the assertion that there exist $\lambda_i \in \Delta(M_i(\mu))$ satisfying (21).

Assume such λ_i exist and define $f_{ij} = B_i \lambda_{ij}$. The row sums of f are B_i , and (21) says that its column sums are p_j . Define

$$\beta_i = \max_j \frac{p_j}{d_{ij}}.$$

Because p_j is proportional to e^{μ_j} , the set $M_i(\mu)$ is exactly the set of chores maximizing p_j/d_{ij} . Hence $f_{ij} > 0$ implies $j \in M_i(\mu)$ and therefore $p_j = \beta_i d_{ij}$. The complementarity certificate of Theorem 4.1 applies.

Conversely, let (x, p) be a competitive equilibrium, and let $f_{ij} = p_j x_{ij}$. The row and column equations imply

$$\sum_{j=1}^m p_j = \sum_{j=1}^m \sum_{i=1}^n f_{ij} = \sum_{i=1}^n B_i = B.$$

Set $\mu_j = \log p_j$ and $\lambda_{ij} = f_{ij}/B_i$. Then each λ_i is a probability vector, and by Lemma 3.1 it is supported on the maximizers of p_j/d_{ij} , equivalently on $M_i(\mu)$. The flow column sums give

$$\sum_i B_i \lambda_{ij} = \sum_i f_{ij} = p_j.$$

Since $\sum_j p_j = B$, the normalization formula (20) holds. Therefore (19) holds. $\square$

Corollary 6.2 (Stationarity verification for fixed prices). *Given μ , stationarity of Φ can be checked by a linear feasibility problem: send supply B_i from each agent i to chores in $M_i(\mu)$ so that chore j receives demand p_j from (20). The hard problem is not checking a proposed active set $M_i(\mu)$, but finding a μ for which this balancing flow exists.*

7 Promise versions and exact recovery

The preceding results immediately imply exact algorithms under additional information. These promise results are not solutions of the unrestricted open problem, but they clarify what kind of information would be sufficient.

Corollary 7.1 (Active-set oracle). *If an oracle returns a graph G with $S(f) \subseteq G \subseteq T(p, \beta)$ for some equilibrium certificate, then an exact equilibrium and utility profile can be recovered in polynomial time.*

Proof. Run $\text{LP}(G)$ and apply Proposition 5.1. □

Corollary 7.2 (Nondegenerate active-set identification). *Suppose an equilibrium certificate (p, β, f) has support $S = S(f)$ and a strict separation gap*

$$\gamma = \min_{(i,j) \notin S} \left(\beta_i - \frac{p_j}{d_{ij}} \right) > 0.$$

If an approximation procedure returns sufficiently accurate approximations to p and β to identify exactly the equalities $p_j = \beta_i d_{ij}$ on S and the strict inequalities off S , then exact recovery follows in polynomial time by solving $\text{LP}(S)$.

Proof. Under the stated identification guarantee, the recovered graph is the true support S and satisfies $S \subseteq T(p, \beta)$. Proposition 5.1 applies. □

Remark 7.3. *The gap assumption in Corollary 7.2 is a genuine promise. In degenerate instances the gap can vanish because there may be tight zero-flow edges. Even when it is positive, it can be too small for polynomial-bit approximate information to identify the active set. Therefore this corollary is not a resolution of the original exact-computation question.*

8 What is and is not proved

The results above prove that the unrestricted exact problem has a compact algebraic certificate and that, once the active graph is known, exact recovery is linear programming. They also show that the nonsmooth potential (17) has no spurious stationary points: every stationary point is a competitive equilibrium and every competitive equilibrium gives a stationary point after normalization.

They do not prove a polynomial-time exact algorithm for the unrestricted Brânzei–Sandomirskiy problem. The missing step is the following.

Find, in time polynomial in n , m , and the bit length of the rational input, a graph G for which $\text{LP}(G)$ has positive optimum, or equivalently find an exact stationary point of (17) together with its active sets, without enumerating exponentially many graphs.

This is precisely where the certificate stops. The stationary formulation repackages the same obstruction: for a fixed μ , the balancing condition is a flow problem, but locating an exact μ whose induced active sets balance the normalized price vector is the unresolved active-set-selection problem.

Acknowledgment of status

This manuscript should be cited, if at all, as partial progress only. It gives a rigorous reformulation, exact certificate, and conditional exact recovery theorem. It does not solve the unrestricted polynomial-time computation problem.

References

- [1] S. Brânzei and F. Sandomirskiy. Algorithms for competitive division of chores. *Mathematics of Operations Research*, 49(1):398–429, 2024. doi:[10.1287/moor.2023.1361](https://doi.org/10.1287/moor.2023.1361); arXiv:[1907.01766](https://arxiv.org/abs/1907.01766).
- [2] R. T. Rockafellar. *Convex Analysis*. Princeton University Press, 1970.
- [3] A. Schrijver. *Theory of Linear and Integer Programming*. Wiley, 1986.